ENGINEERING PORTFOLIO · 2026

Muhammet Mert Erdoğan

Mechanical engineer working on design, structural analysis and manufacturing. BSc Mechanical Engineering, thesis MSc in progress at Istanbul University-Cerrahpaşa. Certified SOLIDWORKS Professional (CSWP).

mmert1erdogan@gmail.com linkedin.com/in/mmerterdogan mmerterdogan.github.io

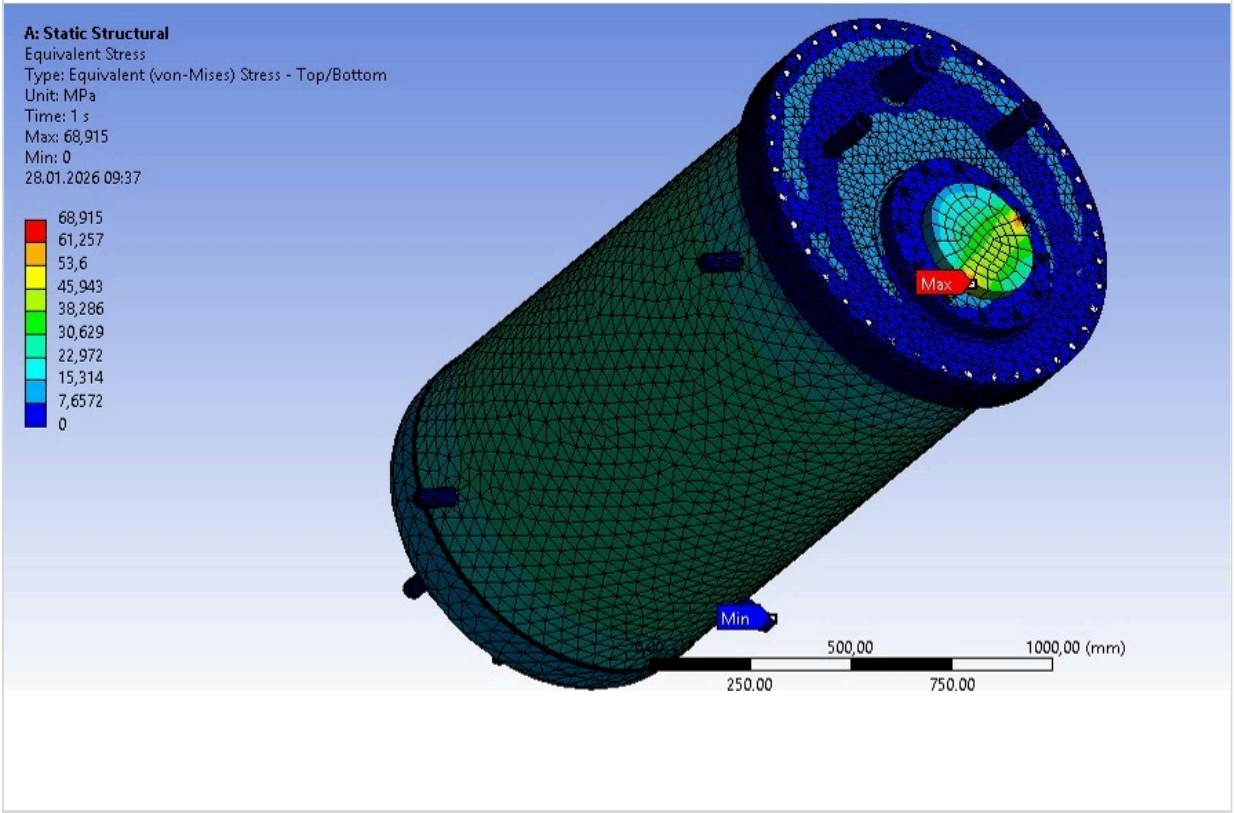
01 · PRESSURE VESSEL · STRUCTURAL ANALYSIS

Where is the design really governed?

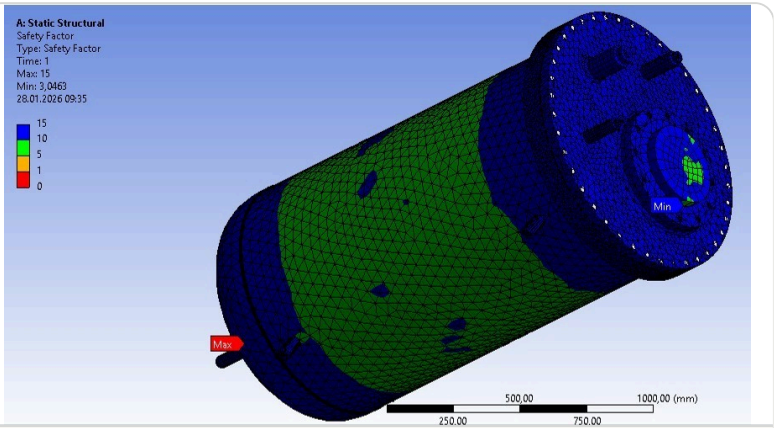
Mechanical System Design project, İÜC · Oct 2025 - Jan 2026
Advisor: Dr. Murat Bozkurt · individual work

Shell theory gave the vessel a comfortable margin. The ANSYS model of the real head and flange showed where that margin is actually lost.

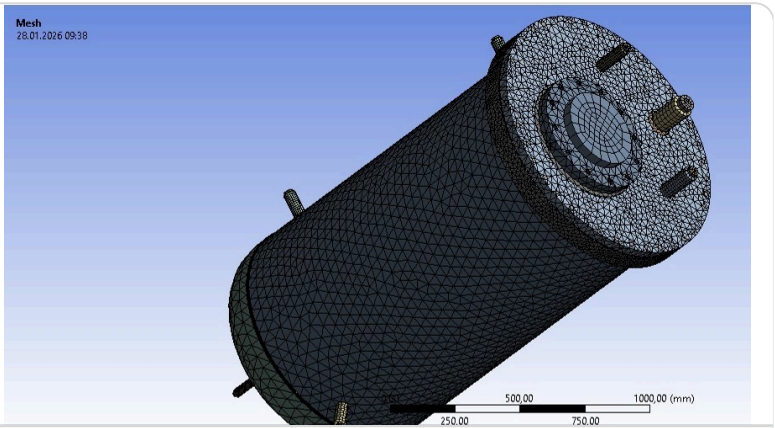
11.2	3.05	68.9 MPa	0.597 MPa
safety factor, shell membrane (theory)	minimum safety factor (FEA)	peak von Mises, at the head	internal pressure, SA-516 Gr.70



Equivalent stress: the peak sits at the head and flange, not in the shell



Safety factor map, minimum 3.05



Mesh, refined at the discontinuities

Two routes, one honest comparison

PROBLEM

Shell r = 450 mm, t = 10 mm, head 12 mm, L = 1600 mm, SA-516 Gr.70 (yield 260 MPa), internal pressure 0.597 MPa.

APPROACH

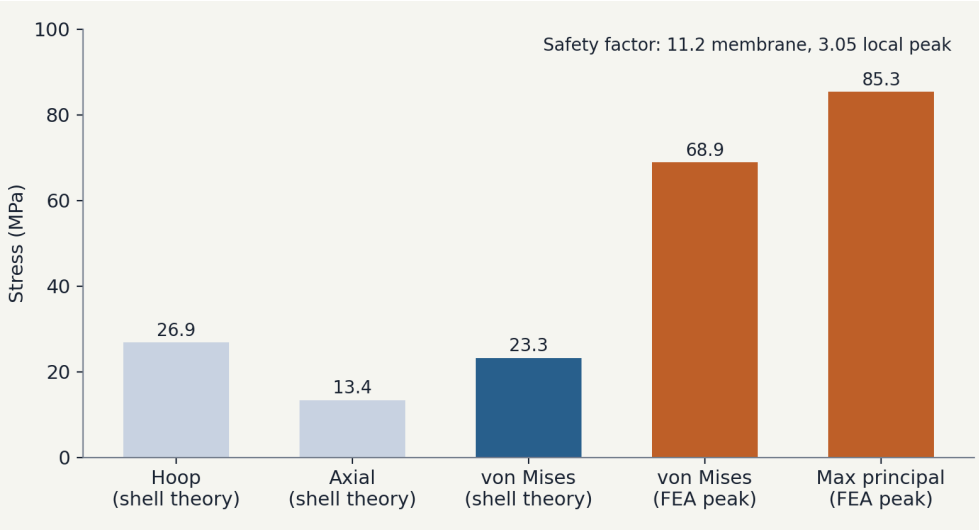
- **Hand calculation:** thin-wall shell theory for hoop, axial and von Mises stress (23.3 MPa) and radial growth (0.051 mm).
- **FEA:** linear static model of the vessel and its bolted head in ANSYS Mechanical 2022 R1.
- **Comparison:** theory describes the shell away from the ends, FEA peaks include bending at discontinuities, so I compared them at two levels.

RESULT

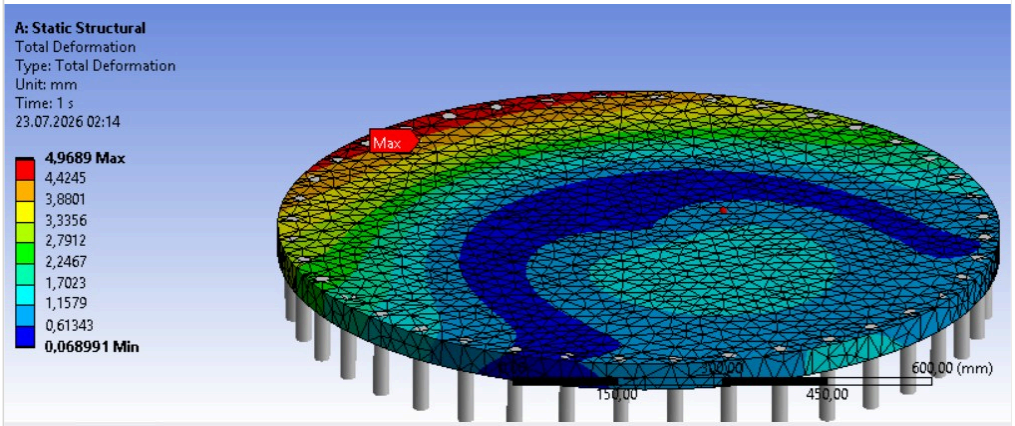
The shell is far from yield, but the head and flange govern the design with a safety factor of **3.05** (260 / 85.3 MPa peak principal stress).

NEXT

Mesh independence at the peak and a minimum-constraint support instead of the fixed base.



Shell theory against FEA peak values



Bolted head model, total deformation

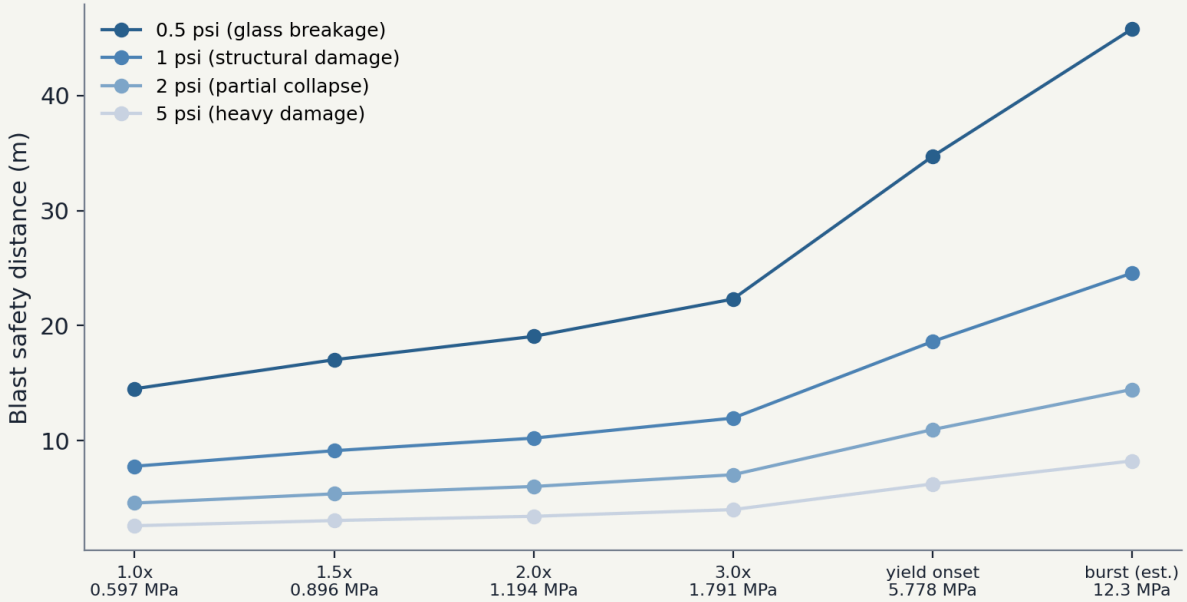
01 · PRESSURE VESSEL · BURST CONSEQUENCE

From stress to a safety distance

Article in preparation with Dr. Murat Bozkurt

A safe stress level does not say how far people should stand if the vessel ruptures. I added a consequence chain to the same vessel.

0.81 MJ	0.19 kg	14.5 m	45.8 m
stored energy at working pressure	TNT equivalent	0.5 psi distance, working pressure	0.5 psi distance, burst case



Blast safety distance by pressure scenario and overpressure threshold (V = 1.1 m³)

METHOD

- **Stored energy:** isentropic expansion of an ideal gas (k = 1.4).
- **TNT equivalent:** 4.184 MJ/kg, with 4.68 MJ/kg as the uncertainty bound.
- **Distance:** Hopkinson-Cranz scaling with Kinney-Graham overpressure values.
- **Scenarios:** 1x to 3x working pressure, yield onset and an estimated 12.3 MPa burst.

> Academic pre-engineering assessment. Fragment throw was discussed qualitatively because the data to calculate it was not available.

02 · TEKNOFEST UAV · DESIGN AND MANUFACTURING

A skid landing gear that survives the landing

TEKNOFEST International UAV Competition · Aug 2024 - Aug 2025
Team project · my part: landing gear

A 7 kg X-frame quadcopter needed a landing gear that is light, keeps the camera clear and absorbs a hard landing. I designed it.

50%
below the specification weight target

7.05 kg
take-off mass of the UAV

292 mm
landing gear height

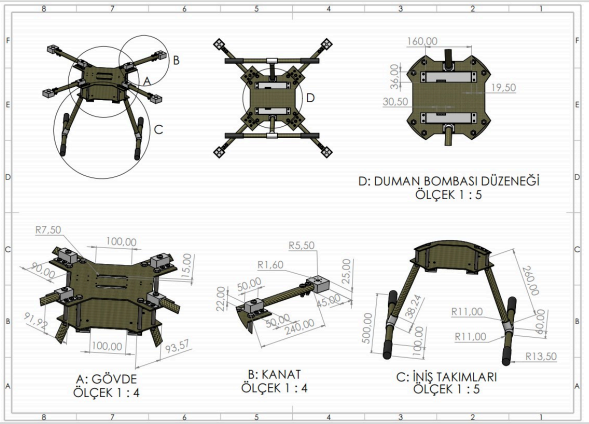
415 mm
landing gear width



Full UAV assembly in SolidWorks



Skid legs: carbon tubes, printed T-joints, rubber pads



Drawing sheet with frame, arm and gear dimensions

Light, stiff and easy to replace

DESIGN DECISIONS

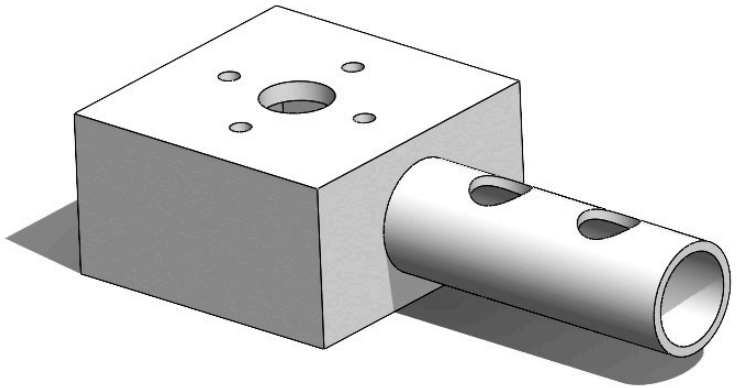
- **Skid gear** instead of arm-mounted legs: keeps it out of the camera's view and spreads the landing load over a larger area.
- **Carbon-fibre tubes** with a 5 mm wall on the legs, thicker than the arms, from a local manufacturer.
- **3D-printed T-joints and angled flanges** in ABS for strength-to-weight and easy printing.
- **Rubber pads** on the skids to soften the landing shock.
- **Bolted to the frame** like the other parts, for easy assembly.

TOOLS

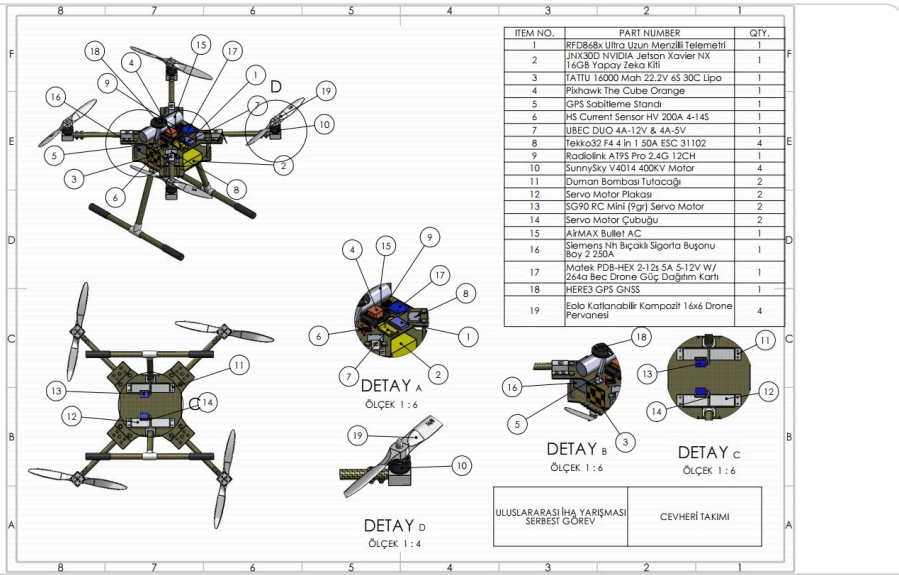
SolidWorks parametric parts, assembly and drawings. ANSYS static structural analysis of the landing gear. FDM printing in PLA, ABS and ASA.

RESULT

The final landing gear came in **50% below the specification weight target**.



3D-printed joint detail



Assembly drawing with bill of materials

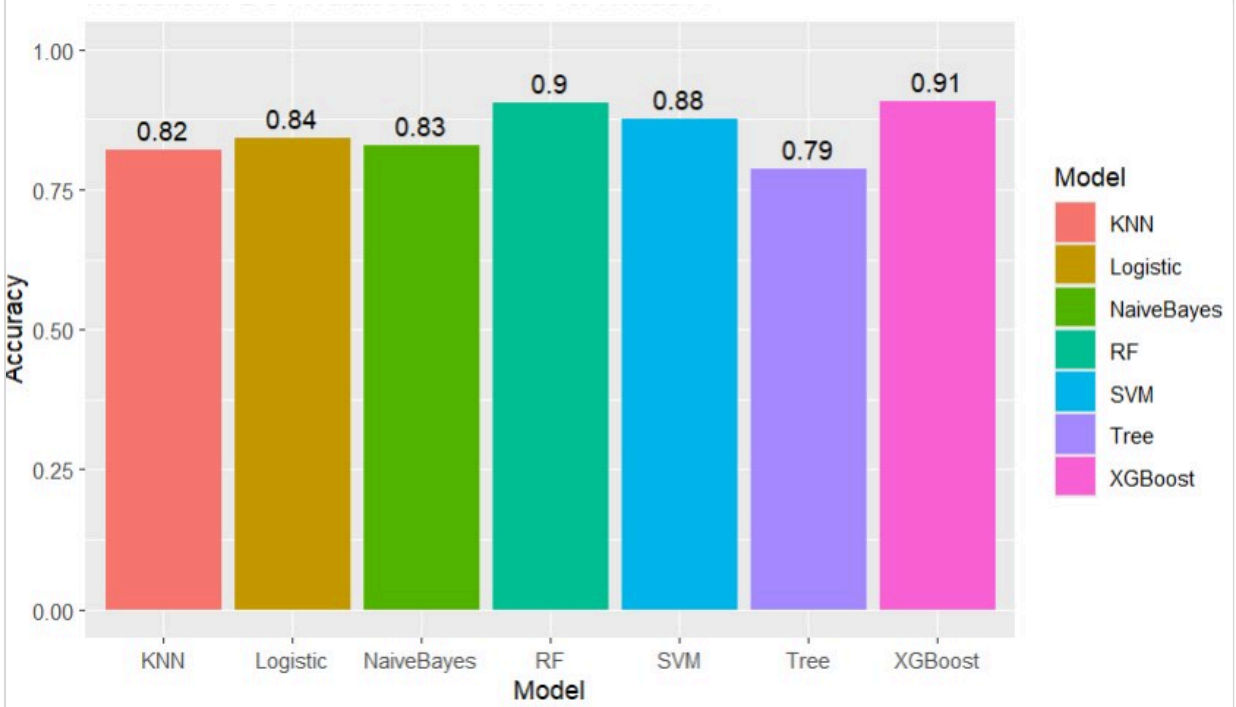
03 · MACHINE LEARNING · QUALITY

Predicting steel plate defects

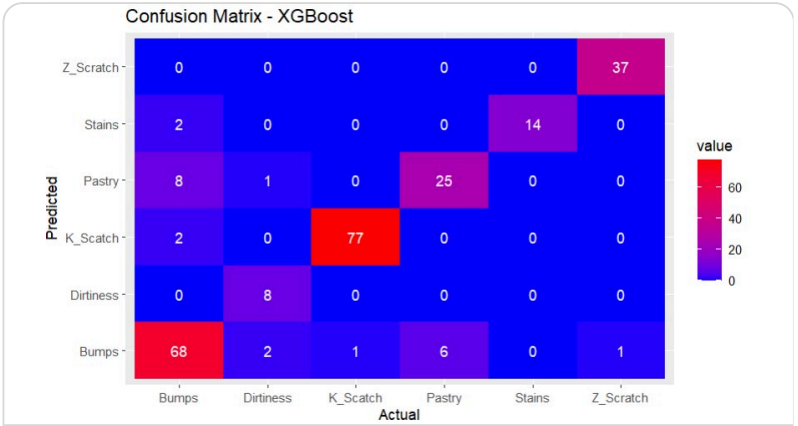
Machine learning course project · Feb 2025 - Jun 2025
Individual work · R

A classifier that tells steel plate fault types apart from inspection measurements, so defective output can be caught and traced back to the process.

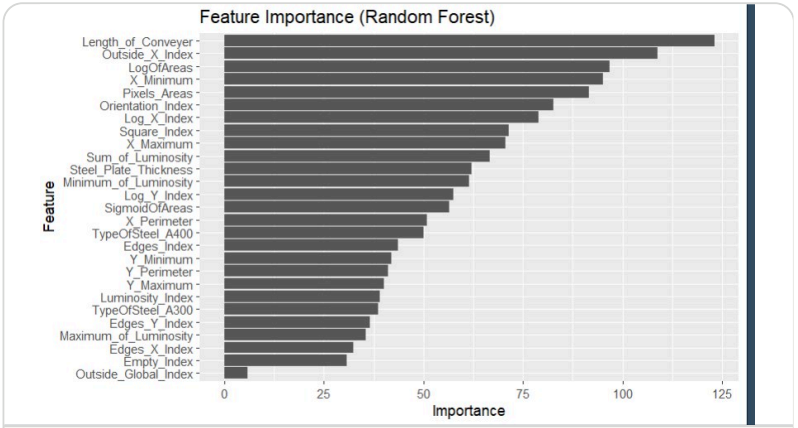
91.3%	0.895	7	27
voting ensemble accuracy	macro F1-score	models compared	input features



Test accuracy of the seven models



Confusion matrix (XGBoost)



Random forest feature importance

From raw measurements to a voting ensemble

DATA AND PREPARATION

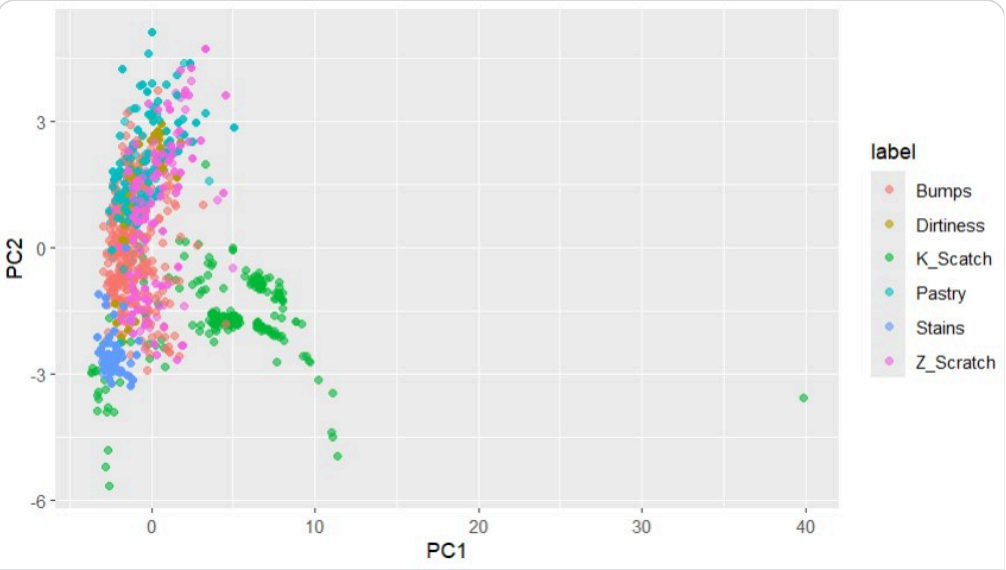
UCI Steel Plates Faults data, 27 geometric and luminosity features per plate. The ambiguous "other faults" class was left out, the data was split 80/20 and scaled, minority classes were oversampled and Boruta selected the informative features.

MODELS

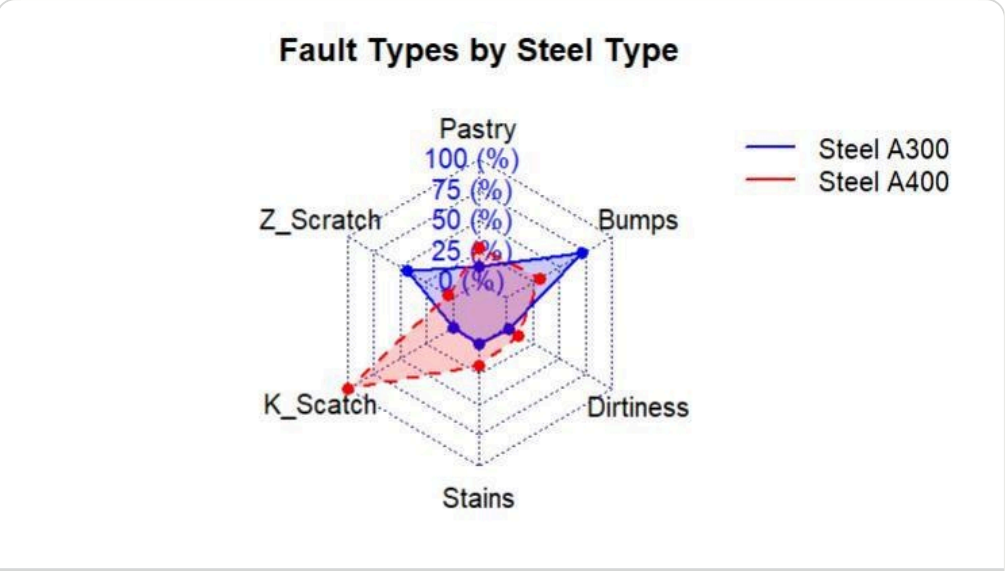
Logistic regression, decision tree, random forest, k-NN, naive Bayes, SVM (5-fold cross-validation for C) and XGBoost. XGBoost (91%), random forest (90%) and SVM (88%) were combined by majority vote: **91.3%**, precision 0.909, recall 0.891.

WHAT IT SAYS ABOUT THE PROCESS

K-scratch and bumps are recognised well, Z-scratch is often confused. Conveyor length ranks highest, so the model leans on line and geometry measurements.



PCA view: K-scratch separates, other faults overlap



Fault profile by steel type (A300 vs A400)

04 · MORE WORK

Formula Student and industry

İÜC RACING FORMULA STUDENT · POWERTRAIN TEAM · 2023 - 2026

Designed the chain guard in SolidWorks and verified its strength with static analysis in ANSYS. Revised the bracket design for a **15% weight reduction**.

INTERNSHIPS

- **FESTO**, 2025: traced mechanical, pneumatic, electrical and software faults in a robotic system and recommissioned it.
- **BEKO**, manufacturing, 2025: found loss sources at 5 critical assembly stations and proposed 2 improvements.
- **Midas Makina**, design, 2024: revised SolidWorks models and prepared manufacturing drawings.
- **Yılmaz Redüktör**, workshop, 2024: machining, heat treatment and gear quality control on site.

SKILLS

SolidWorks (CSWP), ANSYS Mechanical, AutoCAD, Siemens NX, CATIA, Fusion 360, MATLAB/Simulink, R, MS Excel.
Static and modal FEA, topology optimisation, technical drawing and GD&T, design for manufacturability, design for injection moulding, sheet metal design, root cause analysis, lean manufacturing.

CONTACT

mmert1erdogan@gmail.com
linkedin.com/in/mmerterdogan
mmerterdogan.github.io